CSA1708-ARTIFICIAL INTELLIGENCE

LAB EXPERIMENTS

Program 2: write the python program to solve 8.queen problem.

Aim:

To write a Python program to solve the **8-Queens problem** using the **Backtracking**

Algorithm:

1. Start the program.
2. Place one queen in each row.
3. Check whether the position is safe.
4. If safe, place the queen and move to the next row.
5. If no safe position is found, backtrack to the previous row.
6. Repeat until all 8 queens are placed.
7. Display the solution.
8. Stop.

lab 2.py - C:/Users/navit/lab 2.py (3.4.3)

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```
def solve_queens(n):
    board = [-1] * n

    def is_safe(row, col):
        for i in range(row):
            if board[i] == col or abs(board[i] - col) == abs(i - row):
                return False
        return True

    def backtrack(row):
        if row == n:
            return True

        for col in range(n):
            if is_safe(row, col):
                board[row] = col

                if backtrack(row + 1):
                    return True

                board[row] = -1

        return False

    if backtrack(0):
        print("8 Queens Solution:")
        for row in range(n):
            print(" ".join("Q" if board[row] == col else "."
                           for col in range(n)))
    else:
        print("No solution exists.")


solve_queens(8)
```

```
>>>
8 Queens Solution:
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
>>>
```

Result:

Thus, the **8-Queens problem was successfully solved using Python**, and a valid arrangement of 8 queens was obtained.